Entity-Guided Caption Generation for Remote Sensing Images

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***Abstract*—Remote sensing image captioning (RSIC) aims to describe high-level semantics, spatial configurations, and relationships between objects in remote sensing images through natural language. Recent advances have attempted to incorporate entity information from remote sensing images to guide the generation of accurate captions. Although these methods have shown promising results, their effectiveness remains limited due to low entity recognition accuracy and insufficient utilization of the extracted entity information. To address these issues, we propose a dual-stream framework, namely entity-guided caption generation for remote sensing images (EG-Cap). Specifically, the EG-Cap first employs a sequential multientity generator (SMEG) to reformulate entity recognition as a sequence generation task, improving recognition precision by capturing entity contextual dependencies. It then uses an entity–vision adaptive fusion (EVAF) module, via an entity-aware gating mechanism, to adaptively fuse entities with visual features by weighting their semantic compatibility, thereby enhancing the utilization of entity information. Finally, a multiscale contextual aggregation (MCA) module leverages a cascaded Transformer-based decoder to capture multiscale contextual dependencies from the entity–visual representation, ensuring semantically coherent caption generation. Experimental results on the Sydney-Captions, UCM-Captions, and RSICD datasets demonstrate the superior performance of EG-Cap compared to the existing state-of-the-art methods. The implementation code of EG-Cap will be publicly available at: https://github.com/Xiliang-Li/EG-Cap**

***Index Terms*—Entity sequence generation, image caption, multimodal fusion, multiscale aggregation, remote sensing.**

I. Introduction

REMOTE sensing image captioning (RSIC) is a key task in geospatial artificial intelligence that aims to generate human-readable textual descriptions by recognizing salient objects, analyzing their attributes, and modeling semantic relationships. It enables efficient interpretation of complex remote sensing images. With the rapid advancement of aerospace and sensing technologies, the growing availability of high-resolution remote sensing images has heightened the demand for automated semantic understanding. RSIC has thus become increasingly vital for practical applications, including urban planning, disaster assessment, and military reconnaissance [1], [2].

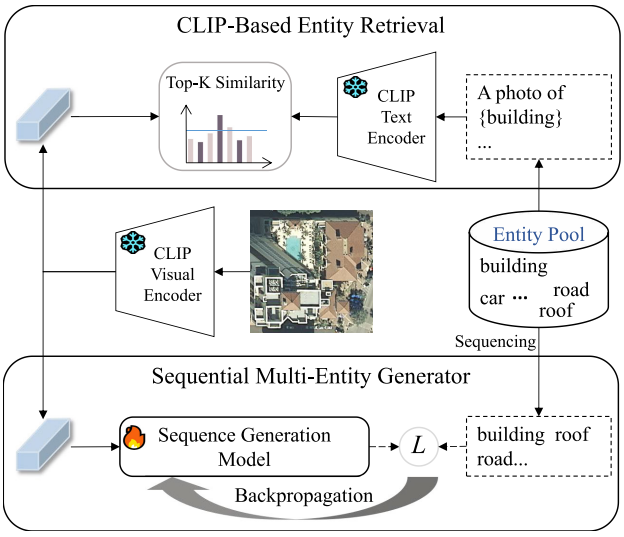


Fig. 1. Comparison between CER and the proposed SMEG. While retrieval-based methods rely on similarity matching against an entity pool composed of entity words from captions, the SMEG introduces sequence-level supervision by modeling discrete entities as outputs, leading to more accurate entity recognition.

Early advances in image captioning have primarily centered on natural images, yielding substantial progress in recent years [3], [4], [5]. These developments laid a methodological foundation for extending captioning techniques to the RSIC domain [6], [7], [8]. However, direct adaptation of these techniques to RSIC tasks often yields suboptimal results due to fundamental differences between natural and remote sensing images, including object scale, texture, and semantic structure [2].

Among recent natural image captioning (NIC) methods, CLIP-based entity retrieval (CER) has emerged as a promising approach [9], [10], [11], [12]. As illustrated in the upper part of Fig. 1, the CER extracts visual features via the pretrained CLIP model [13] and computes their similarity with textual embeddings of predefined entity prompts, selecting top-k candidates as predicted semantic entities. While the CER has demonstrated strong performance in NIC tasks by leveraging entity extraction to enhance semantic alignment between entities and visual representations, its effectiveness in RSIC remains underexplored.

CER-based methods face two key challenges when applied to RSIC tasks: first, remote sensing images typically contain small objects that occupy only a few pixels, resulting in weak and ambiguous visual features, which makes it difficult for

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CLIP to extract meaningful semantic information; second, compared to the natural image domain, which benefits from large-scale training corpora such as the 400 million image–text pairs used for CLIP [13], remote sensing-specific models like RemoteCLIP [14] and GeoRSCLIP [15] are trained on relatively small datasets with only 0.82 million and 5 million pairs, respectively. This limited data scale reduces their generalization ability and hampers the accurate recognition of fine-grained semantic entities in RSIC tasks.

To address these limitations, we propose the sequential multientity generator (SMEG), as shown in Fig. 1. In contrast to CER, which performs isolated entity matching, the SMEG leverages supervision from RSIC datasets through two key mechanisms: 1) transforming discrete entities into ordered sequences to encode semantic relationships and contextualizing small objects to compensate for weak visual features and 2) leveraging sequence supervision to extract richer patterns from limited data, thereby enhancing generalization to fine-grained entities. This design enables precise entity extraction by modeling the inherent semantic structures of remote sensing images.

With the SMEG as the foundational component, we present the complete framework named entity-guided caption generation for remote sensing images (EG-Cap), a dual-stream architecture designed to enhance semantic grounding in caption generation. Specifically, a remote sensing image is first encoded into global visual features and input to the SMEG, which identifies salient entities through sequence-based modeling. These entities are then fused with visual features by the entity–vision adaptive fusion (EVAF) module, which uses a gating mechanism to assign adaptive weights based on semantic compatibility. Finally, the multimodal representations are processed by the multiscale contextual aggregation (MCA) module, which uses a cascaded Transformer-based decoder to capture contextual dependencies and generate coherent captions.

The core contributions of this work are summarized below.

1) A dual-stream framework, EG-Cap, is proposed, integrating a lightweight SMEG as the auxiliary stream. By formulating entity recognition as a sequence generation task, the SMEG module leverages serialized dataset entities as supervisory signals to refine recognition, providing explicit and accurate entity-level guidance for the main captioning stream.
2) An EVAF module is designed with a gating mechanism. Guided by the semantic similarity between entities and visual representations, it dynamically adjusts the contribution of diverse entities to improve utilization efficiency and generate high-quality entity–visual representations.
3) An MCA module is proposed, which employs a cascaded Transformer-based decoder to capture and adaptively integrate multiscale features from fused entity–visual representations via a cascaded Transformer decoder, enabling accurate captions.

To facilitate clarity, the remainder of this article is organized as follows. Section II reviews related work. Section III details the proposed EG-Cap. Section IV presents the experimental results and corresponding analysis. Finally, Section V concludes the article.

II. Related Works

A comprehensive review of relevant studies is provided in this section, mainly categorized into three aspects: NIC, RSIC, and vision–language models (VLMs)/large language models (LLMs) for RSIC.

A. Natural Image Captioning

Early approaches to image captioning primarily relied on retrieval and template-based methods [16]. With the rise of deep neural networks, encoder–decoder frameworks gained widespread adoption, where CNNs encoded visual features and RNNs or LSTMs generated corresponding captions [3], [17], [18]. To improve the ability of the model to focus on important image regions, Xu et al. [19] introduced an attention-based framework for image captioning, allowing the model to focus on salient regions during the decoding process dynamically. Additionally, Lu et al. [20] proposed an adaptive attention mechanism with a visual sentinel, which allows the model to dynamically decide whether to attend to visual features or rely on the linguistic context at each decoding step.

The Transformer [21] architecture mitigated key structural issues inherent in RNN-based models for image captioning, including limited capacity for long-range semantic modeling and restricted parallelism due to sequential token generation. Building on this foundation, Pan et al. [4] proposed the X-Transformer architecture, which integrates X-Linear attention modules to capture high-order interactions between visual and textual features, delivering strong performance on standard image captioning benchmarks. Extending Transformer-based captioning to task-specific scenarios, Cao et al. [22] developed a model for assistive navigation, which improved semantic accuracy by mitigating confounding factors and enhancing the use of discriminative visual details.

With growing interest in vision–language foundation models, VLMs, e.g., Oscar [23] and BLIP-2 [24], and large language models (LLMs), e.g., MiniGPT-v2 [25] and Flan-T5 [26], have further boosted the development of NIC. The former relies on large-scale pretraining on image–text pairs followed by task-specific fine-tuning, enhancing semantic comprehension and the fluency of generated captions. The latter improves its generalization ability and the diversity of generated captions by leveraging the knowledge acquired through massive text learning. However, these methods often incur substantial computational costs. To alleviate this, some studies explored more efficient alternatives by retrieving semantic entities from images and modeling their associations with visual features. This strategy proved effective in NIC [27], [28], and also demonstrated strong potential in zero-shot settings through prompt-based techniques and external knowledge integration [9], [10], [11]. These findings motivated our investigation into entity-aware mechanisms for RSIC.

B. Remote Sensing Image Captioning

The development of NIC [21], [29] laid a foundational technical groundwork for RSIC [2], [6], [30], enabling RSIC

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to achieve rapid advancements in recent years despite its later emergence. However, due to the inherent differences between natural and remote sensing images, the direct application of NIC techniques to RSIC was often limited in effectiveness [31], [32], [33].

In response to this challenge, researchers have proposed domain-specific adaptations of general-purpose techniques, focusing primarily on enhancing multiscale feature modeling and balancing performance with efficiency. For multiscale feature modeling, Liu et al. [34] integrated hierarchical CNN features with an LSTM-enhanced Transformer decoder to capture multiscale information and reduce redundancy; Cheng et al. [35] combined multilevel attention (for multiscale visual features) and contextual attention (for semantic dependencies). For performance–efficiency balance, Meng et al. [36] proposed the RSIC-GMamba, which reduces model complexity while leveraging genetic operations to refine tokens and boost diversity. These approaches collectively enhance RSIC performance through architectural innovations tailored to the unique characteristics of remote sensing images.

Meanwhile, several other studies enhanced the performance of RSIC models by focusing on optimizing fine-grained vision–semantic alignment. Ye et al. [37] proposed a joint-training framework that integrated multilabel semantic supervision with caption generation, enabling shared feature learning to explicitly align visual content with semantic concepts. Wang et al. [38] introduced multilabel classification as an auxiliary task, designing two fusion modules to inject semantic information into the generation process for tighter visual–semantic coupling. Similarly, Wang et al. [12] leveraged the CER method to extract explicit entity information from remote sensing images, reducing the visual–language modality gap and improving alignment-driven captioning performance.

Moreover, in recent years, there has been rapid progress in a distinctive RSIC technology committed to depicting changes across remote sensing images of different periods, which is defined as remote sensing image change captioning (RSICC) [39]. To capture discriminative change features in multitemporal remote sensing images, Liu et al. [40] adopted a transformer-based dual-branch encoder, and a transformer decoder was then used to generate corresponding captions. Meanwhile, Deng et al. [41] broke new ground by proposing ChangeChat, which is the first interactive VLM tailored for remote sensing change analysis. This model not only achieved effective visual–text feature alignment via multimodal instruction tuning but also implemented step-by-step analysis driven by multiturn conversations through chain-of-thought reasoning. Notably, RSICC exhibits substantial differences from natural scene change captioning technologies, mainly due to the need to mitigate visual interference from multiscale objects and seasonal variations with stronger visual impacts [39].

C. VLMs/LLMs for RSIC

In recent years, the advancements in VLMs and LLMs have greatly propelled the development of RSIC. On the one hand, VLMs served as the cornerstone of visual–semantic mapping in RSIC, as they excelled at linking the unique visual characteristics of remote sensing images to basic semantic concepts. For example, Ren et al. [42] adopted BLIP-2 [24] to generate concise descriptions, successfully developing a hybrid-intelligence hierarchical captioning model for remote sensing scenes that inferred spatial relationships between key objects. In a complementary vein, Wang et al. [43] leveraged SAM [44] to extract salient visual regions from remote sensing images. They further parsed noun sequences from captions to guide the integration of fine-grained semantics, thereby enhancing key object representation.

On the other hand, LLMs have improved the text-processing performance of RSIC models through their massive pretrained knowledge and strong language organization abilities [45], [46]. SkyEyeGPT [47] leveraged LLaMA2-Chat [48] to achieve effective region-level descriptions of remote sensing images and videos, and it demonstrated exceptional cross-task generalization capabilities. Lin et al. [49] developed RS-MoE, the first hybrid-expert model for remote sensing, which exhibited outstanding performance across multiple datasets without traditional fine-tuning. In contrast, RS-Agent [50] innovatively integrated LLMs into an agent system and invoked tools such as GeoChat [51], freeing caption generation from the constraints of fixed templates.

While the aforementioned methods have achieved promising results, their performance is constrained by either costly additional annotations or limited accuracy in entity recognition. To address these limitations, this work proposes the EG-Cap, a framework that leverages inherent entity tokens within RSIC datasets as built-in supervision. This design not only enables effective entity extraction but also guides the generation of captions with both efficiency and accuracy.

III. Methodology

This section presents a detailed overview of the proposed EG-Cap framework, as illustrated in Fig. 2. Given a remote sensing image I, visual features are extracted via a pretrained CLIP vision encoder [13]. Specifically, only the `[CLS]` token from the encoder's output is employed as the image representation to facilitate better alignment with entity features in subsequent processing

$$v_I = f_{\text{CLIP}}(I)_{[\text{CLS}]} \tag{1}$$

where $v_I \in \mathbb{R}^{d_1}$ is fed into the SMEG as the initial input, and all three core modules collaborate to generate captions.

First, v_I is input to the SMEG, a lightweight module that reformulates semantic entity recognition as a sequence generation task. The resulting entity sequence is then processed by the EVAF module, which employs an entity-aware gating mechanism to fuse entity and visual features adaptively, thus enhancing the overall semantic representation. Finally, the MCA module utilizes a cascaded Transformer-based decoder network to model multiscale contextual interactions and generate accurate textual descriptions from the fused features.

A. Sequential Multientity Generator

Accurately extracting semantic entities from remote sensing images is critical for high-quality captioning. However,

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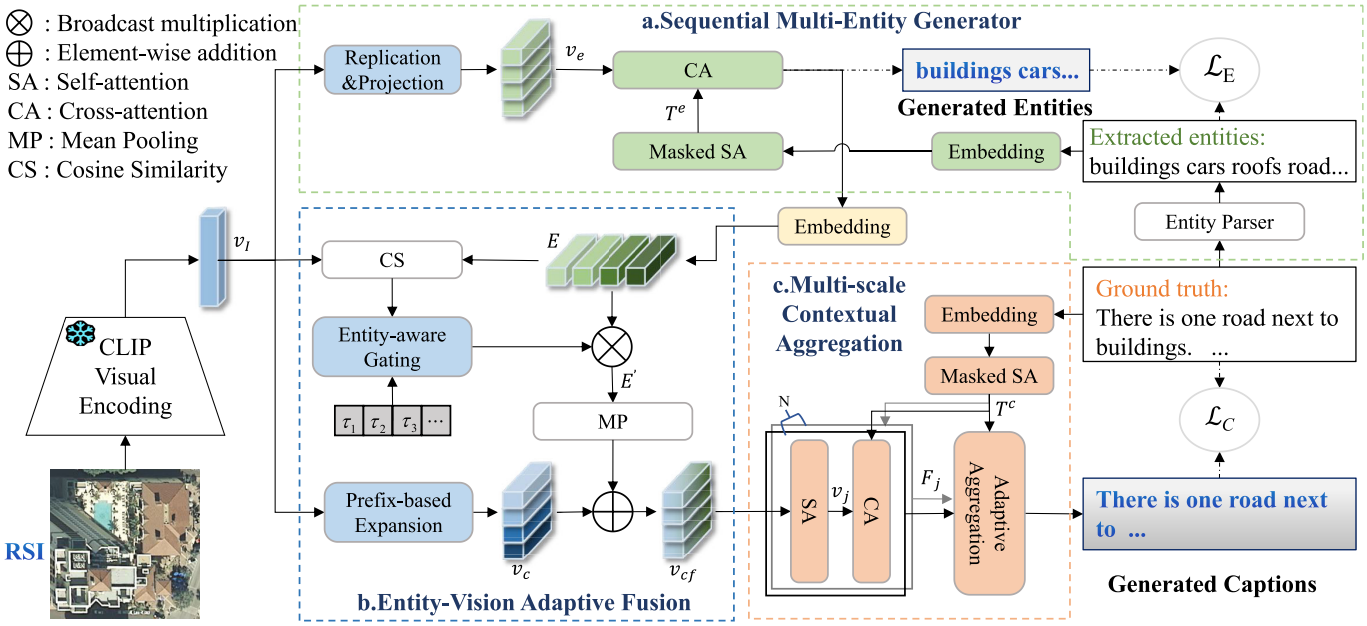


Fig. 2. Overview of the proposed EG-Cap. The EG-Cap comprises three key components: the SMEG, which predicts semantic entities via a sequence generation paradigm; the EVAF module, which employs an entity-aware gating mechanism to integrate entity and visual features adaptively; and the MCA module, which employs a cascaded transformer-based decoder to capture multiscale features and adaptively aggregate these features for accurate caption generation. Note that positional encoding and feed-forward layers are omitted for clarity of illustration.

existing methods often rely on extra annotations to boost accuracy [8], [52]. Inspired by the sequence generation paradigm of image captioning, we propose a low-cost RSIC-specific approach: constructing entity-level supervision from existing RSIC datasets, without additional annotations.

To build this supervision, RSIC caption labels are parsed via NLP tools [53] to extract noun tokens as candidate entities. To reduce redundancy (avoiding repeated mentions of the same entity in different forms) and overspecificity (preventing overly granular terms that hinder generalization), we apply lemmatization to unify word forms. For instance, plurals like "parks" are normalized to "park." These unified entities are serialized into entity sequences, which serve as supervision for the SMEG module. This serialization preserves entity contextual dependencies (e.g., "airport" co-occurs with "runway"), enabling more accurate entity modeling.

To this end, the SMEG module serves as a lightweight auxiliary branch within the EG-Cap framework, tasked with generating an entity sequence from the global visual representation v_I, as shown in Fig. 2(a). To facilitate the decoder's cross-attention (CA) mechanism for text–visual alignment, v_I is first replicated to expand the number of visual units, followed by linear transformation to generate visual priors v_e, formulated as follows:

$$v_e = \text{Repeat}_n(v_I)\, W^e + b^e \tag{2}$$

where $v_e \in \mathbb{R}^{n\times d_2}$, $\text{Repeat}_n(\cdot)$ denotes replicating the input n times along the sequence dimension, $W^e \in \mathbb{R}^{d_1\times d_2}$ is a learnable linear projection matrix, and b^e is the corresponding bias term.

Subsequently, v_e is fed into a single-layer Transformer decoder based on multihead attention (MHA) to perform entity sequence modeling. The MHA splits the input features into multiple attention heads, allowing the model to capture richer and more diverse semantic relationships. The computation is formulated as follows:

$$\text{MHA}(Q, K, V, \text{mask}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^h \tag{3}$$

where $W^h \in \mathbb{R}^{h\cdot d_k\times d_{\text{in}}}$ is a learnable projection matrix, and mask optionally restricts attention by assigning large negative values to disallowed positions before the softmax operation. Each attention head is computed as follows:

$$\begin{aligned}\text{head}_i &= \text{Att}\left(QW_i^Q, KW_i^K, VW_i^V, \text{mask}\right)\\ &= \text{Softmax}\left(\frac{\left(QW_i^Q\right)\left(KW_i^K\right)^\top}{\sqrt{d_k}} + \text{mask}\right) VW_i^V\end{aligned} \tag{4}$$

where $Q, K, V \in \mathbb{R}^{n\times d_{\text{in}}}$ denote the query, key, and value matrices; $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_{\text{in}}\times d_k}$ are the projection parameters for the ith head; $d_k = d_{\text{in}}/h$ is the dimension of each head, and h is the total number of heads.

To effectively extract semantic entity information from the image, the SMEG module leverages v_e to guide the generation of an entity sequence. Specifically, the ground-truth sequence is first transformed into a set of position-aware embeddings $w^e = [w_1^e, w_2^e, \ldots, w_m^e]$, where m denotes the sequence length and each entity embedding is formulated as $w_i^e = e_i + \text{PE}_i \in \mathbb{R}^{d_2}$. Here, e_i refers to the entity embedding of the ith entity token, while PE_i encodes its positional information within the sequence. The embedding sequence w^e is then processed by a multihead self-attention (MSA) module, yielding the context-aware entity representation T^e

$$\begin{aligned}T^e &= \text{MSA}(w^e, \text{mask}^e)\\ &= \text{MHA}(w^e, w^e, w^e, \text{mask}^e)\end{aligned} \tag{5}$$

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where $T^e \in \mathbb{R}^{m\times d_2}$ represents the contextual embeddings, capturing their sequential dependencies. The causal mask $\text{mask}^e \in \mathbb{R}^{m\times m}$ restricts attention to previous positions, preventing information leakage from future tokens. To incorporate visual semantics, T^e is fused with the visual representation v_e via a CA mechanism

$$\begin{aligned} V^e &= \text{CA}\left(T^e, v_e, v_e\right) \\ &= \text{MHA}\left(T^e, v_e, v_e\right) \end{aligned} \tag{6}$$

where $V^e \in \mathbb{R}^{n\times d_2}$ denotes the fused feature. V^e is subsequently passed through a positionwise feed-forward network (FFN) and softmax function to predict the probability distribution over the ith entity token. The resulting entity sequence serves as auxiliary semantic input to the EVAF modules, facilitating entity-aware fusion for enhanced caption generation.

B. Entity–Vision Adaptive Fusion

Entity information encapsulates high-level semantic cues at the word level and plays a critical role in determining the quality of generated captions. Although the SMEG module enhances entity perception via sequence generation, not all predicted entities contribute equally to the RSIC model. This is partly due to the multiscale visual characteristics of entities in remote sensing images, which lead to varied semantic relevance across entities. Therefore, it is essential to achieve an adaptive fusion of visual and entity features while suppressing the influence of irrelevant or erroneous entities to ensure robust caption generation.

To this end, we introduce the EVAF module, as illustrated in Fig. 2(b). The EVAF module evaluates the semantic compatibility between each predicted entity and the global visual context to adaptively select salient entities and modulate their contributions, thereby suppressing noise and accommodating the multiscale characteristics of remote sensing entities. Specifically, the predicted entities are embedded as feature vector $E = [e_1, e_2, \ldots, e_l] \in \mathbb{R}^{l\times d_2}$, and the cosine similarity between each entity feature and the global visual representation, together with an entity-aware gating mechanism, is used to modulate their contributions. The process is formulated as follows:

$$\begin{aligned} s_i &= \sigma\left(\frac{\text{Sim}\left(e_i, v_I W^c\right) - \tau_i}{T_\sigma}\right), \quad i = 1, 2, \ldots, l \\ e_i' &= s_i \cdot e_i \\ E' &= \left[e_1', e_2', \ldots, e_l'\right] \end{aligned} \tag{7}$$

where $W^c \in \mathbb{R}^{d_1\times d_2}$ is a learnable linear projection matrix, $\text{Sim}(\cdot)$ denotes cosine similarity, $\sigma(\cdot)$ is the sigmoid gating function, and τ_i is a learnable threshold for the ith entity, initialized to 0.1, which serves to adaptively regulate the contribution of each entity according to the multiscale characteristics of remote sensing entities. T_σ is a temperature parameter that adjusts gating sharpness, with lower values producing steeper responses.

Since caption generation requires richer visual semantics than entity sequence generation, the EVAF module adopts a more expressive visual feature expansion strategy. Unlike the simple linear broadcasting used in the SMEG, the EVAF leverages a prefix-based expansion approach [54]. Specifically, v_I is first projected into a sequence of visual tokens via a linear layer, concatenated with learnable prefix embeddings, and then processed by a Transformer. The output corresponding to the prefix positions is extracted as the visually conditioned prefix $v_c \in \mathbb{R}^{n\times d_2}$.

To integrate entity-level semantics, E' is aggregated via mean pooling to obtain a global entity representation. This representation is then repeated n times and added to v_c in an elementwise manner, yielding the entity–visual representation

$$v_{cf} = v_c + \text{Repeat}_n\left(\frac{1}{l}\sum_{i=1}^{l} e_i'\right). \tag{8}$$

The entity–visual representation v_{cf}, enriched with entity-aware semantics, is subsequently input into the MCA module for final caption generation.

C. Multiscale Contextual Aggregation

Although the EVAF module enriches visual features with entity semantics, decoding these features into accurate captions remains challenging. We address this by proposing the MCA module, a cascaded Transformer-based decoder that captures multiscale contextual dependencies and uses an adaptive mechanism to fuse features to generate detailed captions.

As illustrated in Fig. 2(c), the caption tokens are first transformed into position-aware embeddings $w^c = [w_1^c, w_2^c, \ldots, w_M^c]$, where M denotes the caption length and each token embedding is defined as $w_i^c = c_i + \text{PE}_i$, with c_i being the word embedding and PE_i the corresponding positional encoding. Following the same encoding procedure as in the SMEG module, w^c is processed by an MSA layer to yield the context-aware caption representation T^c:

$$T^c = \text{MSA}\left(w^c, \text{mask}^c\right) \tag{9}$$

where $T^c \in \mathbb{R}^{M\times d_2}$ denotes the context-aware representations of the input tokens, and $\text{mask}^c \in \mathbb{R}^{M\times M}$ is a causal mask that prevents information flow from future tokens.

The fused visual representation v_{cf} and caption representation T^c are then jointly processed by a cascaded Transformer-based decoder. Each decoder layer first refines the visual features via self-attention (SA), followed by CA with the caption representation. The process is formulated as follows:

$$\begin{aligned} v_j &= \text{SA}\left(v_{j-1}\right) \\ &= \text{MHA}\left(v_{j-1}, v_{j-1}, v_{j-1}\right), \quad v_0 = v_{cf} \\ F_j &= \text{CA}\left(T^c, v_j, v_j\right), \quad j = 1, 2, 3 \end{aligned} \tag{10}$$

where $v_j \in \mathbb{R}^{n\times d_2}$ is the visual representation refined by the jth SA layer, and $F_j \in \mathbb{R}^{M\times d_2}$ denotes the multiscale feature extracted at the jth CA layer (with sequence length M corresponding to caption tokens). Note that though derived from decoder layers of different depths, all F_j share the same feature dimension.

Guided by the context-aware caption representation T^c, the adaptive aggregation module fuses the multiscale features F, as illustrated in Fig. 3. It dynamically estimates the relative importance of each scale through a learnable weighting

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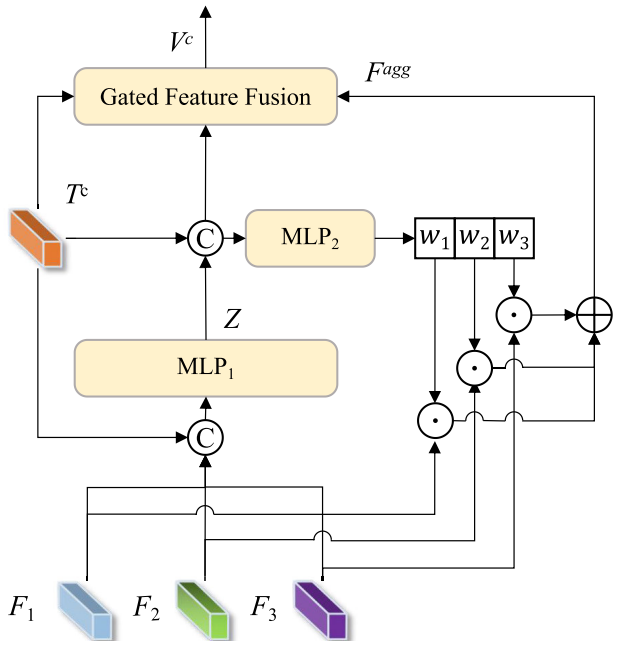


Fig. 3. Structure of the proposed adaptive aggregation module. ⊙ © and denote Hadamard product and feature concatenation, respectively. Feature dimensions are defined as follows: $T^c, Z, F^{\text{agg}}, V^c, F_1, F_2, F_3 \in \mathbb{R}^{M\times d_2}$; $w_1, w_2, w_3 \in \mathbb{R}^{M\times 1}$.

strategy, enabling more effective feature integration. Specifically, two MLPs are employed to first interactively encode concatenated features into an intermediate representation, and then generate normalized fusion weights. The operation is mathematically formulated as follows:

$$\begin{aligned} Z &= \text{MLP}_1\left(\text{Concat}\left(T^c, F_1, F_2, F_3\right)\right) \in \mathbb{R}^{M\times d_2} \\ w &= \text{Softmax}\left(\text{MLP}_2\left(\text{Concat}\left(T^c, Z\right)\right)\right) \\ &= [w_1, w_2, w_3] \in \mathbb{R}^{M\times 3} \\ F^{\text{agg}} &= \sum_{j=1}^{3} w_j \cdot F_j \in \mathbb{R}^{M\times d_2} \end{aligned} \tag{11}$$

where both MLP$_1$ and MLP$_2$ share a similar architecture, each consisting of two linear layers with a GELU activation in between. After that, a gated network is used to further balance the contributions of F^{agg} and T^c. This process is formulated as follows:

$$\begin{aligned} g &= \sigma\left(\text{MLP}_{\text{gate}}\left(\text{Concat}\left(T^c, Z\right)\right)\right) \in \mathbb{R}^{M\times 1} \\ V^c &= \text{LN}\left(g \cdot F^{\text{agg}} + (1-g)\cdot T^c\right) \end{aligned} \tag{12}$$

where MLP$_{\text{gate}}$ consists of a linear layer, LayerNorm, GELU, and another linear layer applied sequentially. The sigmoid activation σ produces the gating vector $g \in \mathbb{R}^{M\times 1}$, which assigns tokenwise weights to balance the contributions of fused and contextual features.

Analogous to the modeling of entity sequences, the output representation V^c is processed by a positionwise FFN followed by a softmax activation to estimate the probability distribution over the next word in the caption.

D. Optimization and Training Objectives

To jointly supervise the entity sequence generation and image captioning tasks, we adopt a cross-entropy loss for each. Let $y^e = \{y_1^e, y_2^e, \ldots, y_{t^e}^e\}$ and $y^c = \{y_1^c, y_2^c, \ldots, y_{t^c}^c\}$ denote the ground-truth sequences of lengths t^e (entity) and t^c (caption), respectively. At each time step t, the model outputs a probability distribution $p_t^e \in \mathbb{R}^{k^e}$ and $p_t^c \in \mathbb{R}^{k^c}$ over the entity and caption vocabularies of size k^e and k^c, respectively. The total training loss is defined as follows:

$$\begin{aligned} \mathcal{L}_{\text{total}} &= \lambda_e \mathcal{L}_E + \mathcal{L}_C \\ &= -\lambda_e \sum_{t=1}^{t^e}\sum_{k=1}^{k^e} y_{t,k}^e \log\left(p_{t,k}^e\right) - \sum_{t=1}^{t^c}\sum_{k=1}^{k^c} y_{t,k}^c \log\left(p_{t,k}^c\right) \end{aligned} \tag{13}$$

where λ_e is scheduled using a cosine annealing strategy following [55]. t^e and t^c denote the lengths of the entity and caption sequences, respectively, while k^e and k^c represent the corresponding vocabulary sizes. $p_{t,k}^e$ and $p_{t,k}^c$ are the predicted probabilities of selecting the kth word at time step t for the entity and caption sequences. According to the ground truth, $y_{t,k}^e = 1$ (or $y_{t,k}^c = 1$) if the kth word is the correct token at position t; otherwise, the value is 0.

After training the EG-Cap with the cross-entropy loss, we follow prior works [33], [36] to further fine-tune it using the self-critical sequence training (SCST) method [56], which directly optimizes nondifferentiable metrics such as CIDEr-D using policy gradient. Specifically, the SCST loss is formulated as follows:

$$\mathcal{L}_{\text{SCST}} = -\frac{1}{N}\sum_{i=1}^{N}(r_i - \bar{r}_i)\cdot \log p_i \tag{14}$$

where N is the batch size, r_i denotes the reward (CIDEr-D score is used in this work) of the sampled caption for the ith image, $\bar{r}_i$ is the baseline reward computed as the average of beam search results, and $\log p_i$ represents the log-probability of the sampled caption.

Since the SCST mainly enhances the fluency and coherence of generated language, it contributes little to entity sequence modeling. Therefore, during this stage, the SMEG module is frozen, and only the caption generation module is updated using ground-truth captions.

IV. Experiments

A. Datasets and Evaluation Metrics

1) Datasets: To evaluate the effectiveness of the proposed EG-Cap, experiments are conducted on three widely used RSIC datasets: RSICD [2], Sydney-Captions [6], and UCM-Captions [6]. In line with prior state-of-the-art approaches [33], [34], [36], [57], [58], each dataset is randomly split into training, validation, and testing subsets using a ratio of 8:1:1. For a fair comparison, we adopt the same random splitting strategy and perform five independent runs for each dataset. The best and worst results from the five runs are discarded, and the mean and standard deviation of the remaining three runs are reported. The three datasets are briefly introduced as follows.

RSICD [2] is a large-scale benchmark dataset for RSIC, comprising 10 921 images with a resolution of 224 × 224. Five manually annotated captions are provided for each image, resulting in 54 605 textual descriptions in total. The dataset

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spans 30 distinct scene categories, such as residential areas, farmland, and rivers.

Sydney-Captions [6] consists of 613 aerial images collected from the Sydney metropolitan area, each annotated with five captions, yielding a total of 3065 sentences. The images have a resolution of 500 × 500 and a spatial resolution of 0.5 m per pixel, covering 7 scene categories, such as airport, park, and residential areas.

UCM-Captions [6] comprises 2100 aerial images with a resolution of 256 × 256 and a spatial resolution of approximately 0.3 m per pixel. Each image is annotated with five descriptive captions, resulting in 10 500 descriptions across 21 scene categories such as forest, parking lot, and overpass.

2) Evaluation Metrics: To objectively assess the quality and reliability of the generated captions, five widely used evaluation metrics in image captioning are employed: BLEU-N ($N = 1, 2, 3, 4$) [59], ROUGE-L [60], METEOR [61], CIDEr-D [62], and SPICE [63].

BLEU-N is used to measure n-gram precision by comparing generated captions with reference sentences, focusing on exact token-level matches. ROUGE-L is applied to evaluate the longest common subsequence, highlighting recall-oriented alignment. METEOR is computed by aligning words based on stemming and synonym matching, combining both precision and recall. CIDEr-D is calculated using TF-IDF weighted n-grams to quantify both consensus with reference captions and diversity of generated expressions. SPICE is designed to evaluate semantic content by parsing captions into scene graphs and comparing semantic tuples such as objects, attributes, and relationships.

In addition, we report a composite metric S_m^*, defined as the average of BLEU-4, METEOR, ROUGE-L, and CIDEr-D, which jointly reflects fluency, lexical overlap, and semantic consistency.

B. Implementation Details

During feature extraction, the visual features are obtained from a CLIP model (ViT-B/32 backbone) [13], pretrained on the RS5M dataset [15]. Following prior work [17], [33], the final layer of the vision encoder is removed, and the `[CLS]` token is extracted from the penultimate layer, yielding a 768-D feature vector. This feature is then projected into a 512-D space, consistent with the dimension of the word embeddings (i.e., $d_1 = 768$, $d_2 = 512$).

The EVAF module uses two sequential SA layers with the number of attention heads $h = 8$ as part of a prefix-based expansion strategy, while the MCA module incorporates three SA layers with $h = 4$. All other MHA layers in the framework are configured with $h = 8$. Entity sequences and captions share the same tokenization and embedding strategy. However, the caption vocabulary excludes infrequent words (i.e., those occurring fewer than five times), whereas no such filtering is applied to the entity vocabulary. The maximum sequence length is set to 20, with a special beginning-of-sequence token `<bos>` and end-of-sequence token `<end>` appended to each sequence. Out-of-vocabulary words are replaced by `<unk>`, and unused positions are padded with `<pad>`.

The model is trained using the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.98$. The initial learning rate is set to 1.0, and a learning rate scheduling strategy with 10 000 warm-up steps is adopted during the cross-entropy training stage. The coefficient λ_e in (13) is annealed from 0.8 to 0 over 20 epochs using a cosine decay schedule, to balance the weight of entity sequence generation loss and caption generation loss. To mitigate overfitting of the SMEG module, λ_e is set to 0 after the 8th training epoch. During SCST training, the learning rate is reset to 5×10^{-6}. The batch size is set to 50, the dropout rate is 0.1, and the hidden layer dimension of the FFN is 2048. The maximum number of training epochs is set to 100. Early stopping is triggered if the CIDEr-D score on the validation set does not improve for five consecutive epochs. In the inference stage, the search width is set to 1 for entity sequence generation and 5 for caption generation. All training and evaluation procedures are conducted on an NVIDIA GeForce RTX 3090 GPU.

C. Quantitative Comparison With State-of-the-Art

We use the refined dataset versions released by Li et al. [7], which have improved annotation consistency due to standardized text formatting. These advantages enhance supervision reliability for entity modeling. For fair comparison (since most prior works use original datasets), quantitative results on both original and refined versions are reported. Subsequent experiments (ablation studies and visualizations) adopt refined datasets to leverage their improved annotation quality.

Although the proposed method does not involve LLMs, this section compares our approach with state-of-the-art non-LLM-based methods and LLM-based counterparts to comprehensively analyze the strengths and weaknesses of different methodologies. Given that LLM-based methods typically exhibit enormous parameter sizes, their performance is reported separately to ensure the objectivity of comparisons. Specifically, non-LLM-based methods and our proposed approach are detailed in Tables I–III, model complexity is shown in Table IV, while LLM-based methods are summarized in Table V. For consistency, all evaluation scores are presented as percentages.

On the UCM-Captions (Table II) and RSICD (Table III) datasets, our method achieves leading performance across most metrics, with gains directly attributed to our core module designs. For UCM-Captions, our method achieves the best performance across all metrics. Specifically, it outperforms RSIC-GMamba by + 12.57 in CIDEr-D (464.22 versus 451.65) and + 4.29 in S_m^* (173.71 versus 169.42), and surpasses MG-Transformer by + 1.90 in SPICE (60.29 versus 58.39). This is because the SMEG module enhances key semantic accuracy via explicit entity recognition, the EVAF module refines semantic representation through EVAF, and the MCA module boosts caption coherence with hierarchical contextual modeling.

For the RSICD dataset, our method demonstrates the best performance in most metrics. When compared to RSIC-GMamba, it gains + 13.57 in CIDEr-D (291.70 versus 278.13), + 2.03 in SPICE (49.46 versus 47.43), and + 4.38 in S_m^* (109.10 versus 104.72). Notably, on RSICD, our method ranks

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TABLE I

QUANTITATIVE COMPARISON OF THE PROPOSED MODEL AGAINST EXISTING METHODS ON THE SYDNEY-CAPTIONS. FOR CLARITY, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDER-D, AND SPICE ARE ABBREVIATED AS B-1, B-2, B-3, B-4, M, R, C, AND S, RESPECTIVELY. THE BEST PERFORMANCE UNDER EACH METRIC IS INDICATED IN BOLD, WHILE THE SECOND-BEST IS DENOTED WITH UNDERLINE. RESULTS ANNOTATED WITH ‡ ARE OBTAINED ON THE REFINED VERSION OF THE DATASET [7]

Methods	B-1	B-2	B-3	B-4	M	R	C	S	S_m^*
mLSTM [6]	54.60	39.50	22.30	21.20	20.50	-	37.20	-	-
Soft-attention [2]	73.22	66.74	62.23	58.20	39.42	71.27	249.93	-	104.71
Hard-attention [2]	75.91	66.10	58.89	52.58	38.98	71.89	218.19	-	95.41
SM-Att+LSTM [58]	74.30	65.35	58.59	51.81	36.41	67.72	234.02	39.76	97.49
Word-Sentence [64]	78.91	70.94	63.17	56.25	41.81	69.22	204.11	-	92.85
GVFGA+LSGA [65]	76.81	68.46	61.45	55.04	38.66	70.30	245.22	45.32	102.31
HCNet [66]	76.86	71.09	65.73	61.02	39.80	71.72	247.14	-	104.92
MLAT [34]	83.23±1.35	77.88±3.99	72.98±4.11	68.29±3.93	43.81±2.11	75.99±2.90	277.24±6.06	-	116.32±2.96
M^2Transformer [67]	82.25±1.65	76.19±1.12	71.04±1.83	66.30±1.83	44.20±1.49	75.21±1.98	275.44±13.65	43.64±1.74	115.29±2.69
PKG-Transformer [57]	83.17±1.02	77.83±2.31	72.84±1.88	68.24±1.39	45.28±0.86	77.06±1.54	284.76±10.84	44.05±1.05	118.83±2.51
ICNET [16]	84.68±0.68	79.85±1.77	75.61±3.47	71.50±4.57	47.40±4.59	77.91±1.30	302.58±8.78	47.28±3.94	124.85±2.04
MG-Transformer [33]	86.68±1.07	80.87±2.47	75.88±3.54	71.22±4.36	47.70±3.11	79.70±3.86	327.76±14.49	48.09±1.93	131.37±5.19
RSIC-GMamba [36]	86.68±0.57	81.65±0.42	77.05±0.89	72.48±1.33	47.81±0.40	80.24±0.23	**343.57±16.73**	**48.44±0.78**	**136.03±3.72**
Ours	**87.12±0.52**	**82.02±0.90**	**77.30±1.26**	**72.90±1.37**	**48.42±0.77**	**81.76±0.97**	339.07±4.51	47.66±2.10	135.54±0.78
Ours‡	88.30±0.62	83.23±0.99	78.64±0.89	74.34±0.73	49.08±1.92	82.05±1.23	347.8±15.69	49.45±1.66	138.32±4.67

TABLE II

QUANTITATIVE COMPARISON OF THE PROPOSED MODEL AGAINST EXISTING METHODS ON THE UCM-CAPTIONS. FOR CLARITY, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDER-D, AND SPICE ARE ABBREVIATED AS B-1, B-2, B-3, B-4, M, R, C, AND S, RESPECTIVELY. THE BEST PERFORMANCE UNDER EACH METRIC IS INDICATED IN BOLD, WHILE THE SECOND-BEST IS DENOTED WITH UNDERLINE. RESULTS ANNOTATED WITH ‡ ARE OBTAINED ON THE REFINED VERSION OF THE DATASET [7]

Methods	B-1	B-2	B-3	B-4	M	R	C	S	S_m^*
mLSTM [6]	63.50	53.20	37.50	21.30	20.30	-	44.50	-	-
Soft-attention [2]	74.54	65.45	58.55	52.50	38.86	72.37	261.24	-	106.24
Hard-attention [2]	81.57	73.12	67.02	61.82	42.63	76.98	299.47	-	120.23
SM-Att+LSTM [58]	81.15	74.18	68.14	62.96	43.54	77.93	338.60	48.75	130.76
Word-Sentence [64]	79.31	72.37	66.71	62.02	43.95	71.32	278.71	-	114.00
GVFGA+LSGA [65]	83.19	76.57	71.03	65.96	44.36	78.45	332.70	48.53	130.37
HCNet [66]	88.26	83.35	78.85	74.49	48.65	83.91	351.83	-	139.72
MLAT [34]	90.35±1.04	86.86±1.31	83.68±1.65	80.77±2.11	54.57±1.61	87.67±1.46	383.15±17.52	-	151.54±5.60
M^2Transformer [67]	88.90±1.32	85.98±1.49	82.74±1.51	80.89±1.36	51.13±2.21	84.45±2.61	418.41±17.27	53.43±2.43	155.97±4.36
PKG-Transformer [57]	90.48±1.15	87.04±1.09	84.10±1.33	81.39±1.67	54.66±2.05	86.57±2.00	427.49±12.37	57.01±2.40	162.53±3.52
ICNET [16]	91.02±0.65	87.52±0.89	84.31±0.96	81.35±1.00	55.25±0.14	87.85±0.15	435.66±3.68	58.18±1.23	165.02±1.16
MG-Transformer [33]	91.28±1.23	88.11±2.29	85.37±2.17	82.95±2.08	56.01±1.66	88.77±2.06	448.39±4.08	58.39±1.32	169.03±2.05
RSIC-GMamba [36]	91.40±1.12	88.42±1.30	85.72±1.43	83.18±1.62	54.65±0.56	88.22±0.47	451.65±6.68	57.17±0.37	169.42±1.15
Ours	**92.13±0.15**	**89.18±0.51**	**86.55±0.29**	**84.07±0.52**	**56.78±0.49**	**89.78±0.55**	**464.22±18.82**	**60.29±1.71**	**173.71±4.73**
Ours‡	93.17±0.45	90.64±0.50	88.47±0.48	86.44±0.52	57.68±1.13	90.51±0.75	470.34±3.75	59.53±0.81	176.24±1.31

second in BLEU-1 (71.74 versus Word-Sentence's 72.40). This is because Word-Sentence leverages a CNN-based word extractor and multilabel loss function to enhance BLEU-1 performance. Our method, in contrast, focuses on holistic semantic integrity, leading to superior performance in more comprehensive metrics (e.g., CIDEr-D and SPICE).

Furthermore, the last two rows in Tables I–III report the performance of the proposed model on both the original and refined versions of the datasets. The results show that using refined datasets leads to consistent performance improvements across all three datasets. To objectively evaluate the effectiveness of different model components, all experiments in Sections IV-D and IV-E are conducted on the refined datasets.

As presented in Table IV, the proposed model achieves a favorable balance between parameter size and computational efficiency. With 60.57 M parameters, it maintains a moderate model size, which is significantly smaller than large-scale models such as MLAT (149.77 M). Furthermore, in terms of inference cost, the proposed model requires

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TABLE III

QUANTITATIVE COMPARISON OF THE PROPOSED MODEL AGAINST EXISTING METHODS ON THE RSICD. FOR CLARITY, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDER-D, AND SPICE ARE ABBREVIATED AS B-1, B-2, B-3, B-4, M, R, C, AND S, RESPECTIVELY. THE BEST PERFORMANCE UNDER EACH METRIC IS INDICATED IN BOLD, WHILE THE SECOND-BEST IS DENOTED WITH UNDERLINE. RESULTS ANNOTATED WITH ‡ ARE OBTAINED ON THE REFINED VERSION OF THE DATASET [7]

Methods	B-1	B-2	B-3	B-4	M	R	C	S	S_m^*
mLSTM [6]	50.57	32.42	23.19	17.46	17.84	35.02	31.61	-	25.48
Soft-attention [2]	67.53	53.08	43.33	36.17	32.55	61.09	196.43	-	81.56
Hard-attention [2]	66.69	51.82	41.64	34.07	32.01	60.84	179.25	-	76.54
SM-Att+LSTM [58]	66.99	55.23	47.03	40.68	32.55	58.02	257.38	46.87	97.16
Word-Sentence [64]	**72.40**	58.61	49.33	42.50	31.97	62.60	206.29	-	85.84
GVFGA+LSGA [65]	67.79	56.00	47.81	41.65	32.85	59.29	260.12	46.83	98.48
MLAT [34]	69.46±1.06	59.20±1.25	51.02±1.32	44.69±1.30	33.95±0.50	61.81±1.22	269.75±3.70	-	102.55±1.39
M^2Transformer [67]	68.44±1.20	56.57±1.25	48.10±1.25	41.56±1.26	32.69±1.01	59.12±0.93	258.58±3.68	45.43±0.92	97.99±1.67
PKG-Transformer [57]	69.67±1.74	58.30±1.39	50.45±1.13	44.31±1.24	33.32±1.31	60.78±1.30	274.01±2.88	46.91±0.79	103.11±1.31
MG-Transformer [33]	70.27±1.54	59.80±0.24	52.08±0.46	46.01±0.65	34.17±0.52	62.27±0.93	285.10±5.01	48.07±0.98	107.89±1.78
RSIC-GMamba [36]	69.38±0.76	59.04±0.11	51.42±0.41	45.38±0.75	33.63±0.40	61.73±0.88	278.13±6.16	47.43±1.05	104.72±1.96
Ours	71.74±0.88	**60.94±0.81**	**52.90±0.78**	**46.67±1.18**	**35.01±0.35**	**63.02±0.54**	**291.70±3.48**	**49.46±0.34**	**109.10±1.38**
Ours‡	72.92±0.76	62.14±0.81	53.93±0.78	47.64±0.80	35.39±0.48	63.70±0.35	296.71±4.25	50.44±0.79	110.86±1.34

TABLE IV

QUANTITATIVE COMPARISON OF MODEL COMPLEXITY IN TERMS OF PARAMETER COUNT AND FLOPS ACROSS THE PROPOSED AND EXISTING METHODS

Metrics	Parameters (M)↓	FLOPs (G)↓
MLAT [34]	149.77	-
MG-Transformer [33]	38.56	1.60
RSIC-GMamba [36]	39.21	1.50
Ours	60.57	1.47

1.47 GFLOPs, which is slightly lower than the RSIC-GMamba (1.50 GFLOPs) and the MG-Transformer (1.60 GFLOPs), demonstrating superior efficiency while preserving competitive representational capacity.

As shown in Table I, on the original Sydney-Captions dataset, our method achieves superior performance across BLEU-1 to BLEU-4, ROUGE-L, and METEOR, which validates the advancement of the proposed approach. However, our method lags behind RSIC-GMamba in CIDEr-D (339.07 versus 343.57) and SPICE (47.66 versus 48.44). This gap arises because RSIC-GMamba uses a state space model (SSM), which efficiently captures long-range dependencies with lower computational complexity. This advantage is especially notable on small datasets like Sydney-Captions. Nevertheless, with respect to the composite score S_m^* (135.54 versus 136.03), the proposed method exhibits merely a marginal performance gap relative to RSIC-GMamba. This finding underscores that the proposed method also delivers commendable performance on small datasets, validating its effectiveness in data-constrained scenarios.

For approaches incorporating LLMs, Table V shows their more prominent performance on large-scale datasets, mainly due to pretrained semantic knowledge and strong language modeling capabilities that fully manifest with sufficient data. For instance, on the RSICD dataset, BITA [17] outperforms our proposed method. This superiority stems from the pretrained LLMs, which provide extensive semantic knowledge and contextual reasoning capabilities, enhancing generalization to better adapt to diverse scenes in large datasets. In contrast, our proposed method achieves more advantages over these LLM-based approaches on small-scale datasets, particularly in ROUGE-L and CIDEr-D metrics. This suggests that while LLMs improve the generalization capability of RSIC models, their performance advantages diminish on small-scale datasets due to insufficient data to leverage their pretrained knowledge.

D. Ablation Studies

To assess the contribution of each proposed component, ablation studies are conducted on the MCA, EVAF, and SMEG modules across the three benchmark datasets, as reported in Table VI.

To establish a baseline, a variant of the model is constructed by removing the three core components: the MCA, EVAF, and SMEG. To ensure model trainability, lightweight replacements are introduced for EVAF and MCA. Specifically, the EVAF module is replaced by a broadcasting-and-mapping strategy, similar to that used in the SMEG module, and the MCA module is substituted with a single-layer Transformer decoder. These simplified alternatives maintain structural compatibility without introducing additional semantic modeling capabilities.

The performance of the baseline model is reported in the first row of each dataset section in Table VI. Given that the ablation study adopts refined datasets, the baseline model already achieves competitive results. When the MCA module is integrated into the baseline (as shown in the second row in Table VI), the model's performance on evaluation metrics is significantly improved. For instance, on the UCM-Captions dataset, the CIDEr score increases by 17.11 and BLEU-4 by 2.16. This improvement confirms that the MCA module, which captures and adaptively fuses multiscale features via its cascaded Transformer-based decoder, enhances both the coherence of generated captions and their relevance to the original annotations.

The EVAF module further enhances the model's capacity to mine visual features. Specifically, as indicated in the third row of the results, when compared to the scenario where

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TABLE V

QUANTITATIVE COMPARISON OF THE PROPOSED MODEL AGAINST EXISTING METHODS LEVERAGING LLMS ON RSIC BENCHMARKS, INCLUDING SYDNEY-CAPTIONS, UCM-CAPTIONS, AND RSICD. FOR CLARITY, BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, CIDER-D, AND SPICE ARE ABBREVIATED AS B-1, B-2, B-3, B-4, M, R, C, AND S, RESPECTIVELY. THE BEST PERFORMANCE UNDER EACH METRIC IS INDICATED IN BOLD, WHILE THE SECOND-BEST IS DENOTED WITH UNDERLINE

Datasets	Methods	B-1	B-2	B-3	B-4	M	R	C	S	S_m^*
Sydney-Captions	SkyEyeGPT [47]	**91.85**	**85.64**	**80.88**	**77.40**	46.62	77.74	181.06	-	95.66
	RS-MoE-7B [49]	84.56	76.10	69.25	63.02	**56.23**	76.62	279.76	-	118.91
	RSGPT [45]	82.26	75.28	68.57	62.23	41.37	74.77	273.08	-	112.86
	RS-CapRet [46]	78.70	70.00	62.80	56.40	38.80	70.70	239.20	-	101.28
	Ours	87.12±0.52	82.02±0.90	77.30±1.26	72.90±1.37	48.42±0.77	**81.76±0.97**	**339.07±4.51**	47.66±2.10	**135.54±0.78**
UCM-Captions	BITA [17]	88.89	83.12	77.30	71.87	46.88	83.76	384.50	54.88	146.75
	SkyEyeGPT [47]	90.71	85.69	81.56	78.41	46.24	79.49	236.75	-	110.22
	RS-MoE-7B [49]	**94.81**	87.09	79.57	72.34	**66.97**	62.74	396.46	-	149.63
	RSGPT [45]	86.12	79.14	72.31	65.74	42.21	78.34	333.23	-	129.88
	RS-CapRet [46]	84.30	77.90	72.20	67.00	47.20	81.70	354.80	-	137.68
	RS-LLaVA [68]	88.70	82.88	77.70	72.84	47.98	85.17	349.43	-	138.86
	Ours	92.13±0.15	**89.18±0.51**	**86.55±0.29**	**84.07±0.52**	56.78±0.49	**89.78±0.55**	**464.22±18.82**	**60.29±1.71**	**173.71±4.73**
RSICD	BITA [17]	77.38	66.54	57.65	50.36	**41.99**	**71.74**	**304.53**	**54.79**	**117.15**
	SkyEyeGPT [47]	**86.71**	**76.66**	**67.31**	**59.99**	35.35	62.63	83.65	-	60.40
	RS-MoE-7B [49]	78.61	63.60	51.44	43.35	39.30	60.77	109.46	-	63.22
	RSGPT [45]	70.32	54.23	44.02	36.83	30.10	53.34	102.94	-	55.80
	RS-CapRet [46]	72.00	59.90	50.60	43.30	37.00	63.30	250.20	-	98.45
	Ours	71.74±0.88	60.94±0.81	52.90±0.78	46.67±1.18	35.01±0.35	63.02±0.54	291.70±3.48	49.46±0.34	109.10±1.38

TABLE VI

ABLATION STUDY ON THE IMPACT OF THE REFINED MCA, EVAF, AND SMEG MODULES ACROSS THE RSICD, SYDNEY-CAPTIONS, AND UCM-CAPTIONS DATASETS

Datasets	MCA	EVAF	SMEG	B-1	B-2	B-3	B-4	M	R	C	S	S_m^*
Sydney-Captions	×	×	×	84.92±0.94	79.71±1.74	74.76±2.22	70.12±2.74	46.87±1.44	79.69±1.59	319.41±11.35	46.59±0.86	129.03±3.98
	✓	×	×	85.26±0.77	80.09±0.79	75.47±0.89	71.19±0.92	47.00±1.97	80.27±1.70	335.63±9.44	47.64±2.41	133.52±3.25
	✓	✓	×	86.74±1.39	81.51±1.40	76.66±1.53	72.16±1.77	47.58±0.88	80.87±1.10	337.44±5.67	49.38±1.84	134.51±2.14
	✓	✓	✓	**88.30±0.62**	**83.23±0.99**	**78.64±0.89**	**74.34±0.73**	**49.08±1.92**	**82.05±1.23**	**347.8±15.69**	**49.45±1.66**	**138.32±4.67**
UCM-Captions	×	×	×	90.74±0.86	87.32±1.29	84.17±1.71	81.32±1.91	54±0.87	87.33±1.71	436.39±13.07	56.86±0.91	164.76±4.30
	✓	×	×	91.44±0.52	88.42±0.80	85.82±1.05	83.48±1.32	56.1±1.33	88.89±0.43	453.5±17.50	58.82±0.32	170.49±4.85
	✓	✓	×	92.34±0.56	89.58±0.78	87.25±1.00	85.16±1.17	57.41±1.07	90.22±0.47	460.61±2.84	59.35±0.64	173.35±1.05
	✓	✓	✓	**93.17±0.45**	**90.64±0.50**	**88.47±0.48**	**86.44±0.52**	**57.68±1.13**	**90.51±0.75**	**470.34±3.75**	**59.53±0.81**	**176.24±1.31**
RSICD	×	×	×	70.28±0.93	59.81±1.06	51.86±1.06	45.67±1.06	33.97±0.57	62.41±0.51	283.21±7.10	49.10±0.50	106.31±2.25
	✓	×	×	70.96±0.48	60.31±0.38	52.16±0.34	45.83±0.35	34.37±0.27	62.7±0.46	290.96±8.99	49.8±0.40	108.46±2.48
	✓	✓	×	72.40±0.84	61.62±1.04	53.44±1.25	47.08±1.41	35.28±0.73	63.32±1.26	293.61±14.57	49.86±1.95	109.82±4.48
	✓	✓	✓	**72.92±0.76**	**62.14±0.81**	**53.93±0.78**	**47.64±0.80**	**35.39±0.48**	**63.7±0.35**	**296.71±4.25**	**50.44±0.79**	**110.86±1.34**

only the MCA module is activated, the EVAF module provides additional boosts to BLEU-3 and BLEU-4: 1.19 and 0.97 on the Sydney-Captions dataset, 1.43 and 1.68 on the UCM-Captions dataset, and 1.28 and 1.25 on the RSICD dataset, respectively. This reflects its effectiveness in emphasizing visual features, thereby enabling more precise multimodal alignment.

The SMEG module provides explicit entity-level semantic guidance to the model. When combined with the previous two modules (full model in the fourth row), it maximizes overall performance, yielding the largest gains in key metrics such as BLEU-4 (+ 2.18 on Sydney-Captions, + 1.28 on UCM-Captions, and + 0.56 on RSICD) and the composite score S_m^* (+ 9.29, + 11.48, and + 4.55 across the three datasets respectively), which validates that recognizing image entities in a sequence generation form can effectively motivate RSIC task performance.

To summarize, the ablation studies confirm that MCA, EVAF, and SMEG modules contribute complementarily to RSIC performance: MCA optimizes feature fusion, EVAF refines visual–entity alignment, and SMEG drives key metric breakthroughs via entity sequence guidance, collectively advancing RSIC quality.

E. Qualitative Analysis via Visualization

To provide a deeper understanding of model behavior beyond quantitative metrics, qualitative analyses are conducted through visualization in this section. Since the proposed model adopts only the `[CLS]` token from CLIP as the visual representation, it lacks direct spatial alignment with

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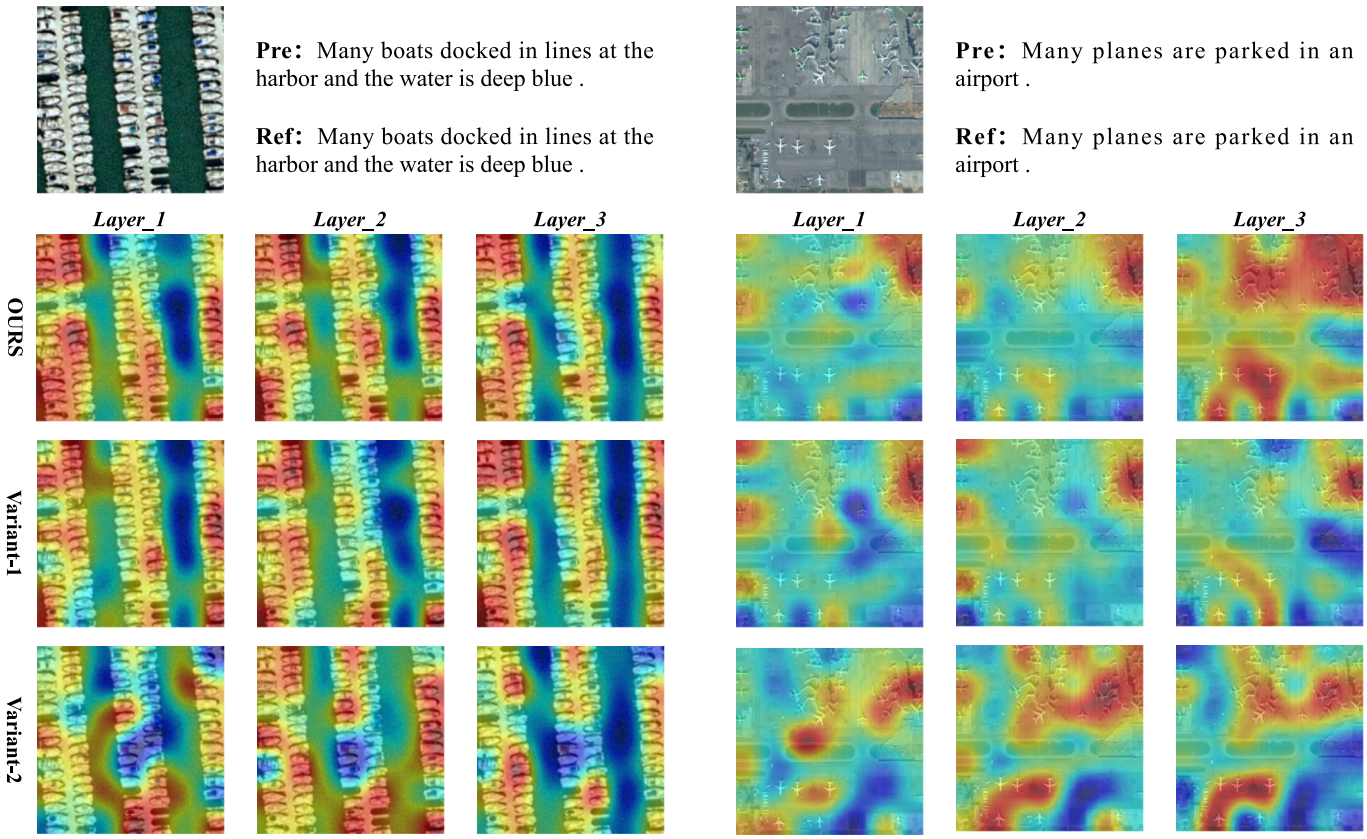


Fig. 4. Comparative attention visualizations derived from the decoder layers of the MCA module, highlighting differences across the full model, variant-1 (without the SMEG), and variant-2 (without the SMEG and EVAF). Heatmaps are superimposed on the original images, where warmer colors red indicate stronger attention responses, and cooler colors blue represent weaker responses.

the input image. To enable spatial visualization, we compute the cosine similarity between the features aggregated by the MCA module and CLIP patch embeddings, approximating the relevance of each image region to the global feature.

As shown in Fig. 4, *Layer_1*, *Layer_2*, and *Layer_3* illustrate the feature visualizations derived from the three CA layers in the MCA module. As the layer depth increases, the model gradually focuses more on key visual entities, such as *boats* and *planes*. In contrast, Variant-1 (without SMEG) captures target entity regions but exhibits less focused and less coherent attention, particularly in deeper layers such as *Layer_3*, where heatmaps show scattered responses. Variant-2 (without SMEG and EVAF) displays even more dispersed attention and weaker semantic alignment; for instance, it fails to consistently highlight *boats* as effectively as Variant-1. These distinctions confirm that SMEG refines attention on key entities while EVAF enhances cross-layer attention consistency, together enabling the model to target semantically important image regions precisely.

To better understand the spatial focus of the model during the prediction of different words, decoder features corresponding to individual tokens are extracted for visualization, as illustrated in Fig. 5. Two representative examples are shown: one depicting a relatively simple scene with a *central building*, and the other involving a more complex geographical context of a *highway*. In both cases, the model successfully captures the visually salient regions that align with the generated captions. For the simple scene, the model accurately attends to the core region, including the circular structure, thereby producing a precise description. In the more complex scenario, although some attention is still allocated to the *highway* when predicting the word *bushes*, the presence of sufficient visual cues related to vegetation enables the model to generate a semantically correct caption.

To further investigate the impact of predicted entity sequences on caption generation, we compare visualization results under two conditions (with and without entity guidance) as shown in Fig. 6. The first row presents two fully accurate caption generation examples: the first demonstrates that proper entity guidance enhances caption precision, while the second shows the model's robustness by generating accurate descriptions even when not all image entities are recognized. The second row's first example illustrates the model's capability to generate long sentences; the second example, where the model incorrectly identifies "airplane" but still produces an accurate caption, indicates the proposed method does not solely rely on SMEG-predicted entities for caption generation. The last row displays two underperformance cases in complex scenarios, verifying the model's limitation: the SMEG module only perceives entity presence in remote sensing images but fails to assess the importance of different entities (e.g., "trees" are of low importance in the relevant scenes).

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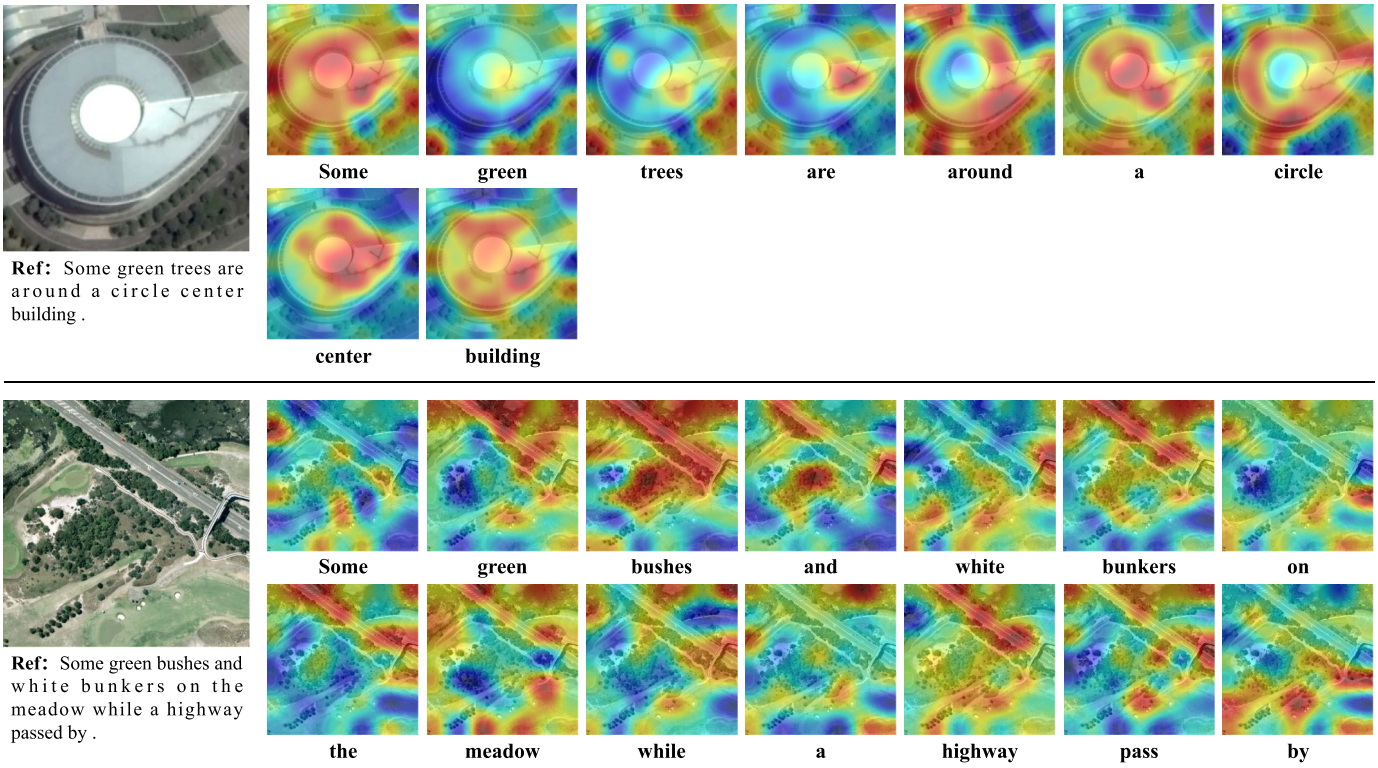


Fig. 5. Visualization of decoder feature heatmaps when predicting different words, superimposed on the original image to reveal attention patterns associated with each word. Warmer colors red indicate stronger attention responses, while cooler colors blue represent weaker responses.



Ref_cap1: This is a dense residential area with houses and plants.
Ref_cap2: It is a dense residential area with lots of houses arranged neatly.
Ref_cap3: There are lots of houses with grey and black roofs arranged neatly.
Ref_cap4: There are lots of houses with grey and black roofs and some plants around them.
Ref_cap5: Lots of houses and plants in the dense residential area.
Ref_entities: area hous plant roof

Full model	**Pre_cap:** Lots of houses and plants in the dense residential area. **Pre_entities:** area hous plant roof
w/o entities	**Pre_cap:** This is a dense residential area with many houses.



Ref_cap1: The bird 's nest stadium is surrounded by vegetation regularly.
Ref_cap2: Red and green football fields are in the middle of the stadium.
Ref_cap3: Some green trees are near a stadium with a baseball field in it.
Ref_cap4: Some green trees are near a stadium with a baseball field in it.
Ref_cap5: The bird 's nest stadium is surrounded by vegetation regularly.

Ref_entities: baseball vegetation stadium field footbal tree

Full model	**Pre_cap:** The bird 's nest stadium is surrounded by vegetation regularly. **Pre_entities:** baseball vegetation stadium field
w/o entities	**Pre_cap:** A baseball field is in a stadium.



Ref_cap1: A villa with grey roofs is surrounded by trees and lawn in the sparse residential area.
Ref_cap2: This is a sparse residential area with a villa surrounded by trees and lawn.
Ref_cap3: a villa with grey roofs and a pond beside is in the sparse residential area.
Ref_cap4: a villa with plants surrounded and a pond beside is in the sparse residential area.
Ref_cap5: a villa with many plants surrounded and a pond beside.
Ref_entities: area tree plant roof villa lawn

Full model	**Pre_cap:** A villa with grey roofs is surrounded by trees and lawn in the sparse residential area. **Pre_entities:** area tree plant roof villa lawn
w/o entities	**Pre_cap:** A villa with plants surrounded is in the sparse residential area .



Ref_cap1: This is an empty airport with some airport buildings beside.
Ref_cap2: An empty airport only with some luggage cars and buildings.
Ref_cap3: An empty airport with some airport buildings beside.
Ref_cap4: This is an airport only with some luggage cars parked on.
Ref_cap5: An empty airport with some airport buildings beside.

Ref_entities: build car park airport luggag

Full model	**Pre_cap:** An empty airport with some airport buildings beside. **Pre_entities:** build park airport airplane
w/o entities	**Pre_cap:** An industrial area with some white buildings and some airport.



Ref_cap1: The round commercial is next to the freeway.
Ref_cap2: At one corner of this four loop flyover a series of high rise buildings is built making up a shape of arc.
Ref_cap3: The round commercial is next to the freeway.
Ref_cap4: in this commercial area a bustling overpass stands.
Ref_cap5: many commercial tall buildings are in two sides of a road with a viaduct.
Ref_entities: loop freeway viaduct round one overpass build area flyover seri road

Full model	**Pre_cap:** Many buildings and some green trees are in a commercial area. **Pre_entities:** build area tree
w/o entities	**Pre_cap:** Some buildings and green trees are near a viaduct.



Ref_cap1: The commercial district is separated by two roads.
Ref_cap2: There is a road passing by a residential area where rows of high rise housings are standing.
Ref_cap3: The commercial district is separated by two roads.
Ref_cap4: Three rows of skyscrapers stands at this prosperous commercial area.
Ref_cap5: Many commercial tall buildings are in a commercial area.
Ref_entities: skyscrap row hous district build area road

Full model	**Pre_cap:** Many buildings and some green trees are in a commercial area. **Pre_entities:** build area tree
w/o entities	**Pre_cap:** Some green trees are in a commercial area.

Fig. 6. Qualitative comparison of caption generation with and without entity guidance. Entities correctly predicted are highlighted in light blue, whereas erroneous predictions are highlighted in red. Note: Ref_cap1–Ref_cap5 denote the reference captions for remote sensing images; Ref_entities represents the entities extracted from Ref_cap1 to Ref_cap5; Pre_cap (Full model) and Pre_cap (w/o entities) refer to the captions generated by the full model and the ablated model with entity information guidance removed, respectively; and pre_entities indicates entities predicted by the SMEG module.

As the order of entities does not affect their role in our model, precision, recall, and $F1$-score are adopted to evaluate the models' entity recognition capability. Table VII reports the performance of SMEG and CER. SMEG starkly

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TABLE VII

PRECISION, RECALL, AND $F1$ SCORES OF SMEG AND CER FOR ENTITY RECOGNITION ON SYDNEY-CAPTIONS, UCM-CAPTIONS, AND RSICD

Datasets	Method	Precision (%)	Recall (%)	F1 (%)
Sydney-Captions	**SMEG**	74.62	85.10	79.52
	CER	39.83	48.10	43.58
UCM-Captions	**SMEG**	88.19	83.52	85.79
	CER	31.67	38.85	34.90
RSICD	**SMEG**	80.08	47.66	59.76
	CER	16.90	17.59	17.23

outperforms CER across all three datasets, demonstrating distinct superiority with precision scores of 74.62 versus 39.83 (Sydney-Captions), 88.19 versus 31.67 (UCM-Captions), and 80.08 versus 16.90 (RSICD). This superiority can be attributed to two primary factors. First, unlike CER, which directly computes the similarity between entity text features and visual features, SMEG employs a sequence generation approach to fully explore the interentity associative relationships. Second, SMEG effectively leverages entity sequences as supervisory information for training. Furthermore, among the three datasets, SMEG attains the optimal performance on UCM-Captions and the lowest on RSICD. Logically, it can be hypothesized that a higher entity count within the dataset would exert a notable negative influence on recognition accuracy. However, our experimental results across Sydney-Captions (39 entities), UCM-Captions (73 entities), and RSICD (408 entities) demonstrate that a smaller entity count does not necessarily correlate with higher recognition accuracy. This difference stems from variations in annotation quality and image sharpness across different datasets.

V. CONCLUSION

This study presents the EG-Cap, a dual-stream framework designed to enhance RSIC performance through explicit entity-level semantic supervision. By reformulating entity recognition as a sequence modeling task via the SMEG module, enabling dynamic multimodal integration through the EVAF, and capturing hierarchical contextual dependencies with the MCA, the EG-Cap establishes a comprehensive pipeline for generating semantically grounded and entity-guided captions. Extensive experiments on three widely used remote sensing captioning benchmarks, namely Sydney-Captions, UCM-Captions, and RSICD, demonstrate that the EG-Cap achieves better performance than existing advanced approaches. For future work, we will refine the alignment between predicted entities and their corresponding visual features, explicitly incorporate entity location positioning mechanisms, and explore joint modeling of local and global cues further to improve the interpretability and contextual consistency of RSIC results.

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